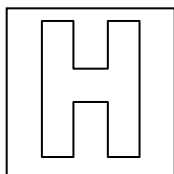


Name:	Index No.:	CT Group: 12
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PIONEER JUNIOR COLLEGE

2013 JC2 Preliminary Examination
HIGHER 1

CHEMISTRY
Paper 2

8872/02

18 September 2013

Candidates to answer Section A on the Question Paper

Additional Materials: Data Booklet
 Writing Paper
 Graph Paper

2 hours

READ THESE INSTRUCTIONS FIRST

Write your name, index number and CT group on all the work you hand in.
Write in dark blue or black pen.
You may use a pencil for any diagrams, graphs or rough working.

Section A

Answer **all** questions.

Section B

Answer any **two** questions on separate answer paper.

You are advised to show all working in calculations.
You may use a calculator.

The number of marks is given in brackets [] at the end of each question or part question.

At the end of the examination, fasten all your work securely together.

FOR EXAMINER'S USE			
Section A		Section B	
1	/ 9	6	/ 20
2	/ 7	7	/ 20
3	/ 4	8	/ 20
4	/ 10	Penalty	s.f. / units
5	/ 10	Total	/ 80

Section A (40 marks)

Answer **all** questions in the spaces provided.

- 1** Calcium ethanedioate, CaC_2O_4 , is a white needle-like crystalline solid. It can be prepared by reacting calcium hydroxide, $\text{Ca}(\text{OH})_2$, and ethanedioic acid, $\text{H}_2\text{C}_2\text{O}_4$.

- (a) (i) Name the reaction between calcium hydroxide, $\text{Ca}(\text{OH})_2$, and ethanedioic acid, $\text{H}_2\text{C}_2\text{O}_4$.

.....

- (ii) Write an equation for the reaction.

.....

[2]

- (b) When a pure sample of solid anhydrous CaC_2O_4 was heated strongly at 400°C , a white solid **B** and 0.028 g of carbon monoxide gas were obtained as the **only products**.

- (i) Given that 1 mole of CaC_2O_4 decomposes to give 1 mole of $\text{CO}(\text{g})$, identify solid **B** and write an equation for the reaction.

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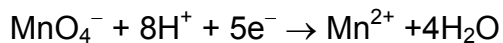
.....

- (ii) Hence, determine the mass of **B** obtained in the reaction.

[4]

- (c) In an experiment, 25.0 cm^3 of $0.050 \text{ mol dm}^{-3} \text{ K}_2\text{C}_2\text{O}_4$ was titrated against $0.022 \text{ mol dm}^{-3}$ acidified KMnO_4 . It was found that 22.70 cm^3 of KMnO_4 was required for complete reaction. During the titration, effervescence was observed and the gas evolved gave a white precipitate in limewater.

The reaction of manganate(VII) in acidic medium is given below.



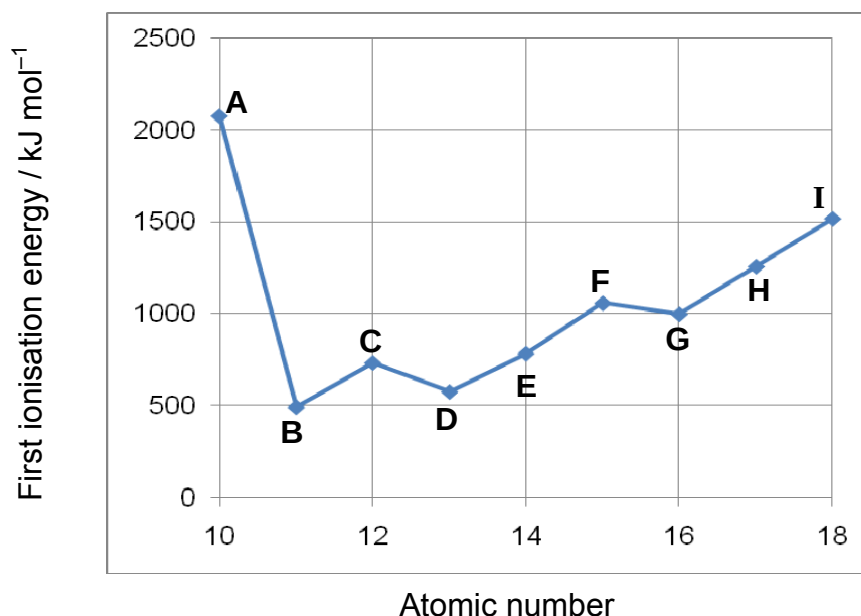
- (i) Using the information above, calculate the mole ratio of $\text{C}_2\text{O}_4^{2-}$ to MnO_4^- in the above reaction.

- (ii) Hence, write an equation for the reaction between $\text{C}_2\text{O}_4^{2-}$ and MnO_4^- in acidic medium.

.....
[3]

[Total: 9]

- 2 The diagram below shows a plot of first ionisation energy against atomic number for the elements **A** to **I**. (The letters are not the chemical symbols for the elements concerned.)



- (a) Give the full electronic configuration of the element **D**.

.....
[1]

- (b) Explain briefly why the first ionisation energy of **G** is less than **F**.

.....
.....
.....
[2]

- (c) (i) Draw a dot-and-cross diagram to show the electron arrangement in the oxide of **D**.

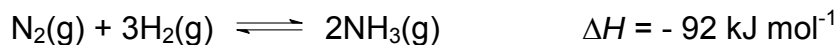
- (ii) The reaction of **D** with chlorine produces a chloride with a M_r of 133.5. Draw a diagram to illustrate the shape of the chloride of **D**. Indicate clearly the bond angle.

- (iii) In terms of structure and bonding, explain why the oxide of **D** has a higher melting point than the chloride of **D**.

.....
.....
.....
.....
[4]

[Total: 7]

- 3** Le Chatelier's Principle predicts that the highest equilibrium yield of ammonia in the Haber process,



should occur at high pressure and at low temperature. However, in practice, these conditions are not used.

- (a)** What are the typical values of pressure and temperature used industrially? Explain why these conditions are used industrially.

.....

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.....

[3]

- (b)** Why are the gases passed through a tower packed with lumps of iron?

.....

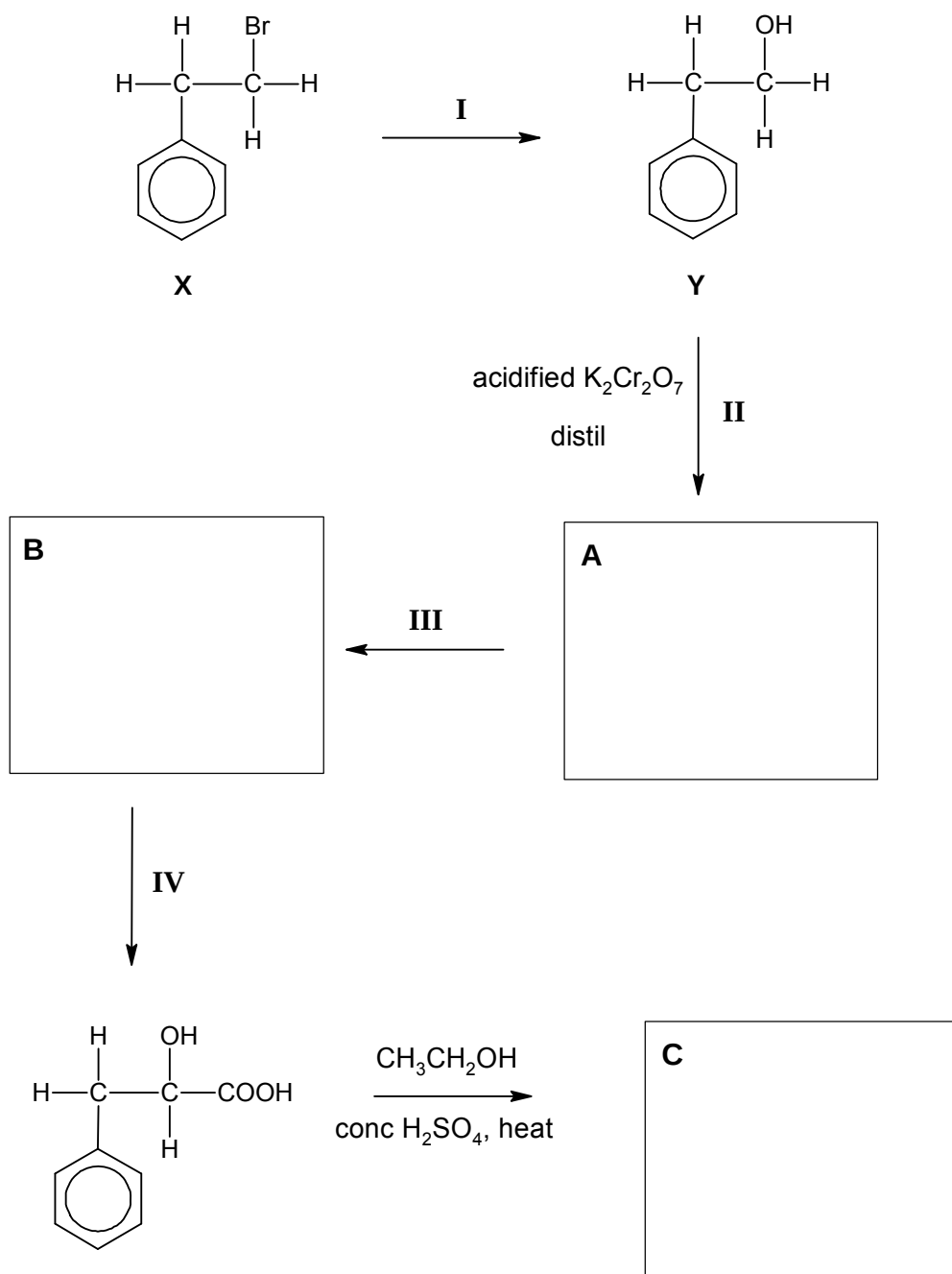
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[1]

[Total: 4]

4 (a) Consider the following reaction scheme.



Draw the structures of the compounds **A**, **B** and **C** in the boxes provided.

[3]

(b) State the type of reaction taking place in steps **I** and **III**.

Step **I**:

Step **III**:

[2]

- (c) State the reagents and conditions required for steps I, III and IV.

Step I:

Step III:

Step IV:

[3]

- (d) Suggest a simple chemical test to distinguish compound X from compound Y. You should state clearly the observations for each compound.

.....

.....

.....

.....

.....

[2]

[Total: 10]

- 5 Compound K has the molecular formula $C_4H_8O_2$.

K decolourises aqueous bromine.

- (a) State the functional group in K that decolourises aqueous bromine.

.....

[1]

- (b) Draw the structural formula of **all** the possible isomers of K which have the following features.

- K decolourises aqueous bromine.
- K reacts with sodium metal but not with aqueous sodium hydroxide.
- No oxygen atom is attached to sp^2 hybridised carbon atom.
- No carbon atom has more than one oxygen atom joined to it.

[3]

- (c) Compound **K** has another isomer, **J**, which does not decolourise aqueous bromine but react with sodium hydroxide at room temperature.

(i) Draw the displayed formula of **J**.

(ii) Give the structure of the organic compound formed when **J** reacts with sodium hydroxide at room temperature.

[2]

- (d) An aqueous solution of an acid HA of concentration $0.100 \text{ mol dm}^{-3}$ has a *pH* of 2.9.

(i) Define the term in italics.

.....

(ii) Using the information, calculate the K_a of HA.

[4]

[Total: 10]

Section B (40 marks)

Answer **two** questions from this section on separate answer paper.

- 6 (a) 5.34 g of a salt, M_2SO_4 (where **M** is a metal), was dissolved in water. The sulfate ion was precipitated by adding excess barium chloride solution. 4.66 g of barium sulfate precipitate was obtained.

- (i) Calculate the number of moles of sulfate ions precipitated.
- (ii) Determine the relative formula mass of M_2SO_4 . Hence, calculate the relative atomic mass of metal **M** and state its identity.

[5]

- (b) State and explain whether metal **M** or calcium would have a higher melting point.

[3]

- (c) State another two physical properties of metal **M**.

[2]

- (d) Do you expect the chloride of **M** to dissolve in water? Briefly explain your answer.

[1]

- (e) (i) When metal **M** reacts with water, effervescence is observed and an alkaline solution is obtained. Write an equation of the reaction.

- (ii) Explain what is meant by the term *standard enthalpy change of neutralisation*.

- (iii) 40 cm³ of 3.0 mol dm⁻³ hydrochloric acid was added to 60 cm³ of 1.4 mol dm⁻³ of the metal **M** hydroxide in a polystyrene cup. The maximum temperature rise recorded was 11.6 °C.

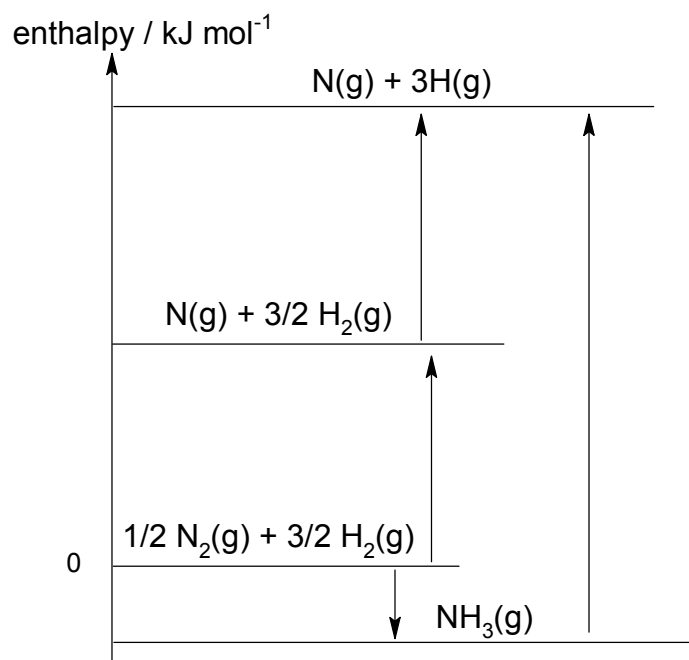
Given that the specific heat capacity of the solution = 4.2 J g⁻¹ K⁻¹ and the process is 100% efficient, calculate the enthalpy change of neutralisation for the reaction.

- (iv) The enthalpy change of neutralisation of ammonia and hydrochloric acid is – 55.2 kJ mol⁻¹.

Account for the difference in the values of the enthalpy change of neutralisation.

[7]

- (f) The enthalpy change of formation of ammonia gas is -46 kJ mol^{-1} . Using the following energy level diagram and other data from the *Data Booklet*, calculate the bond energy of N–H bond in ammonia.

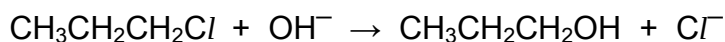


[2]

[Total: 20]

- 7 (a) Halogenoalkanes are useful intermediates in organic synthesis. They can undergo hydrolysis with a base to produce alcohols.

The kinetics of the hydrolysis of 1-chloropropane with aqueous potassium hydroxide may be investigated by determining the concentration of 1-chloropropane remaining at different time intervals. The reaction is represented by the equation below.



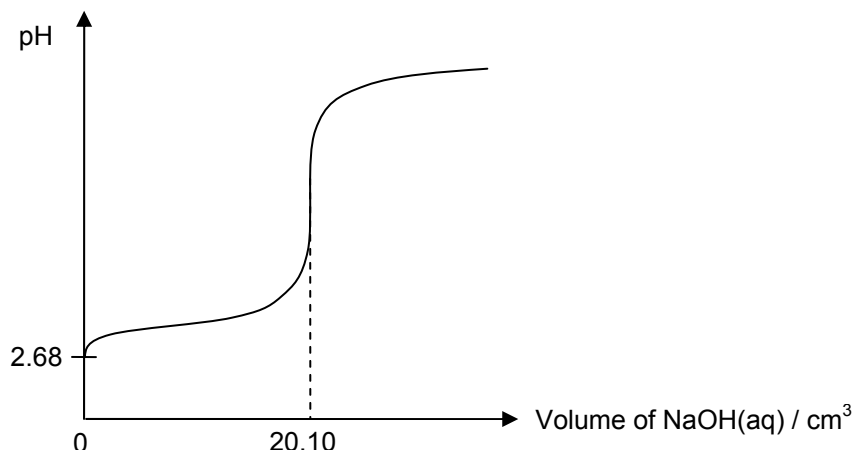
The following results were obtained from two experiments on such a hydrolysis. The reaction was followed twice with different concentrations of potassium hydroxide.

Time / minutes	Experiment 1 [KOH] = 0.10 mol dm ⁻³	Experiment 2 [KOH] = 0.15 mol dm ⁻³
	[1-chloropropane] / mol dm ⁻³	[1-chloropropane] / mol dm ⁻³
0	0.0100	0.0100
40	0.0079	0.0070
80	0.0062	0.0049
120	0.0049	0.0034
160	0.0038	0.0024
200	0.0030	0.0017
240	0.0024	0.0012

- (i) Using the same axes, plot graphs of [1-chloropropane] against time for the two experiments, labelling each graph clearly.
- (ii) Use your graph to deduce the order of reaction with respect to 1-chloropropane. Show your working clearly.
- (iii) Use your graph to determine the initial rate of each reaction. Hence, use it to deduce the order of reaction with respect to potassium hydroxide.
- (iv) Construct a rate equation for the reaction and use it to calculate the rate constant, giving its units.
- (v) Explain, with an aid of the Maxwell-Boltzmann distribution curve, the effect of increasing temperature on the rate of the reaction.

[10]

- (b) Acrylic acid, $\text{CH}_2=\text{CHCOOH}$, is a monobasic acid. 25.0 cm^3 of the acid required 20.10 cm^3 of 0.10 mol dm^{-3} NaOH(aq) for complete reaction, as shown by the following titration curve.



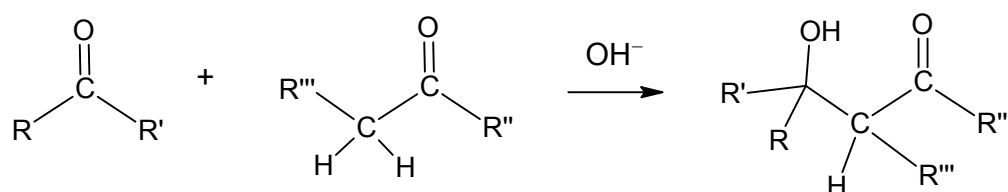
- (i) Use the titration curve to calculate the concentration of acrylic acid used.
- (ii) Using the pH value of the acrylic acid shown in the graph, calculate the concentration of the hydrogen ions in the solution of acrylic acid. Hence, deduce whether acrylic acid is a weak or strong acid.
- (iii) Suggest, with reason, a suitable indicator for the titration of acrylic acid with sodium hydroxide.
- (iv) Before the end-point is reached, the pH of the solution changes gradually as shown in the graph. This is due to the presence of a mixture of acrylic acid ($\text{CH}_2=\text{CHCO}_2\text{H}$) and its salt ($\text{CH}_2=\text{CHCO}_2\text{Na}$) in the solution.

By means of an ionic equation, show how this mixture of acrylic acid and its salt removes the small amount of sodium hydroxide added as the titration proceeds.

[6]

- (c) The aldol reaction is a useful method for making new carbon–carbon bonds in organic chemistry when carbonyl compounds react under suitable conditions.

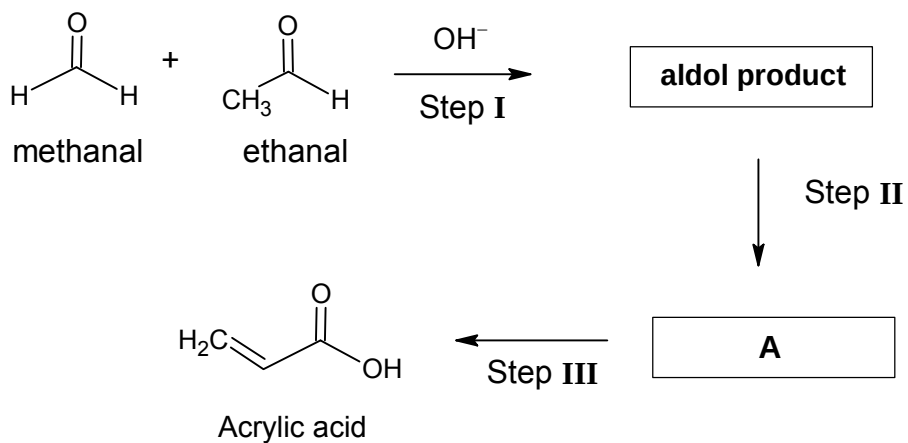
In alkaline medium, two carbonyl compounds undergo aldol reaction as shown below.



[R groups can be alkyl, aryl or H]

- (i) Draw the structure of the aldol product formed between methanal and ethanal.

(ii) Acrylic acid can be synthesised from the following reaction scheme.



Give the structure of compound **A** formed and state the reagents and conditions for Step II and III.

[4]

[Total: 20]

- 8 (a) The chlorides of elements sodium to phosphorus all dissolve or react with water.

Describe the reaction of chlorides of aluminium and phosphorus with water. Suggest the pH values of the resulting solutions and explain your answer with equations. [5]

- (b) (i) Explain what is meant by *first ionisation energy* of an element, **X**.
(ii) State and explain the difference in first ionisation energy between
I aluminium and phosphorus
II magnesium and aluminium [5]

- (c) Compound **P** is found to contain 75.3% carbon, 10.6% hydrogen and 14.1% oxygen by mass. The relative molecular mass of **P** is 112.

When **P** is oxidised with acidified potassium manganate(VII), a compound **Q**, $C_4H_6O_3$, is produced together with propanoic acid. Treatment of **P** with phosphorus pentachloride produces no white fumes.

Q produces a pale yellow precipitate when warmed with alkaline solution of iodine, accompanied by a solution which, when treated with dilute hydrochloric acid, gave compound **R**.

R may also be obtained when $NC - CH_2 - CN$ is heated with a dilute acid. **R** reacts with excess pentachloride to produce a compound **S**, together with white fumes evolved.

Deduce the structure of compounds **P**, **Q**, **R** and **S** and explain the chemistry of the reactions described above. Write equations where appropriate. [10]

[Total: 20]

End of Paper